Junseo Lee – *Curriculum Vitae*

David E. and Stacey L. Goel Quantum Science and Engineering Building
Harvard Quantum Initiative, 60 Oxford St., Rm. 410, Cambridge, MA 02138, USA

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Website: harris-junseo-lee.github.io

Research Interests

Quantum Information and Computation; Quantum Many-Body Systems; Computational Complexity; Statistical Learning

Education

Harvard University, Kenneth C. Griffin Graduate School of Arts and Sciences (GSAS) — Cambridge, MA
Ph.D. Student in Quantum Science and Engineering — Sep. 2026 – Present

- Advisors: Prof. Anurag Anshu and Prof. Sitan Chen (Computer Science); Prof. Jordan Cotler (Physics)
- Affiliations: Harvard Quantum Initiative; Theory of Computation Group
- Supported by the Kwanjeong Ph.D. Fellowship

Yonsei University — Seoul, Korea
B.S. in Electrical and Electronic Engineering — Mar. 2019 – Feb. 2023

- Supported by the Hyundai Motor CMK Science and Technology Scholarship

Chungnam Science High School — Gongju, Korea
Concentration in Mathematics; Early Honors Graduate — Mar. 2017 – Dec. 2018

Research Positions

NSF AI Institute for Artificial Intelligence and Fundamental Interactions — Cambridge, MA
Junior Investigator (Nominator: Prof. Jordan Cotler) — May. 2026 – Present

Institute of Computer Technology, Seoul National University — Seoul, Korea
Research Fellow (Host: Dr. Kabgyun Jeong) — Apr. 2026 – Aug. 2026

Research Institute of Mathematics, Seoul National University — Seoul, Korea
Research Affiliate (Host: Dr. Kabgyun Jeong) — Jan. 2023 – Mar. 2026
Undergraduate Research Assistant (Advisor: Dr. Kabgyun Jeong) — Mar. 2020 – Dec. 2022

Republic of Korea Army (Mandatory Military Service) — Seoul, Korea
Research Scientist (Technical Research Personnel) — Mar. 2023 – Mar. 2026

Publications (Google Scholar, arXiv)

*Equal contribution. †Authors listed alphabetically.

(19) "Hamiltonian locality testing and certification do not achieve the Heisenberg limit,"
Francisco Escudero Gutiérrez, Junseo Lee†, and Sebastian Zur,
Manuscript (2026). arXiv: 2610.03205.

(18) "Optimal purity estimation with incoherent measurements,"
Junseo Lee†, and Chirag Wadhwa,
Manuscript (2026). arXiv: 2609.40248.

(17) "Emergent prethermal symmetries for scalable Hamiltonian learning,"
Myeongjin Shin, Junseo Lee, Iman Marvian, and Yu Tong,
Manuscript (2026). arXiv: 2609.27181.

(16) "Heisenberg-limited Hamiltonian learning without short-time control,"
Myeongjin Shin, Junseo Lee, and Changhun Oh,
PRX Quantum 7, 043011 (2026) DOI: 10.1103/kjvf-jgqj. arXiv: 2604.27838
Long talk at the *26th Asian Quantum Information Science Conference* **(AQIS 2026)**.

(15) "Certifying and learning local quantum Hamiltonians,"
Andreas Bluhm, Matthias C. Caro, Francisco Escudero Gutiérrez, Junseo Lee†, Aadil Oufkir, Cambyse Rouzé, and Myeongjin Shin,
Manuscript (2026). arXiv: 2603.29809.
Contributed talk at the *21st Conference on the Theory of Quantum Computation, Communication and Cryptography* **(TQC 2026)**.

(14) "Optimal certification of constant-local Hamiltonians,"
Junseo Lee†, and Myeongjin Shin,
Manuscript (2025). arXiv: 2512.09778.

(13) "Efficient learning of bosonic Gaussian unitaries,"
Marco Fanizza, Vishnu Iyer, Junseo Lee†, Antonio Anna Mele, and Francesco Anna Mele,
Manuscript (2025). arXiv: 2510.05531.
Contributed talk at the *29th Annual Conference on Quantum Information Processing* **(QIP 2026)**.

(12) "Bounding quantum uncommon information with quantum neural estimators,"
Donghwa Ji, Junseo Lee, Myeongjin Shin, IlKwon Sohn, and Kabgyun Jeong,
Quantum Science and Technology **11**, 015001 (2026). DOI: 10.1088/2058-9565/ae18f4. arXiv: 2507.06091.

(11) "Collapses in quantum-classical probabilistically checkable proofs and the quantum polynomial hierarchy,"
Kartik Anand, Kabgyun Jeong, and Junseo Lee†,
Manuscript (2025). arXiv: 2506.19792.

(10) "Resource-efficient algorithm for estimating the trace of quantum state powers,"
Myeongjin Shin*, Junseo Lee*, Seungwoo Lee, and Kabgyun Jeong,
Quantum **9**, 1832 (2025). DOI: 10.22331/q-2025-08-27-1832. arXiv: 2408.00314.

(9) "Mutual information maximizing quantum generative adversarial networks,"
Mingyu Lee, Myeongjin Shin, Junseo Lee, and Kabgyun Jeong,
Scientific Reports **15**, 32835 (2025). DOI: 10.1038/s41598-025-18476-y. arXiv: 2309.01363.

(8) "Disentanglement provides a unified estimation for quantum entropies and distance measures,"
Myeongjin Shin*, Seungwoo Lee*, Junseo Lee*, Donghwa Ji, Hyeonjun Yeo, and Kabgyun Jeong,
Physical Review A **110**, 062418 (2024). DOI: 10.1103/PhysRevA.110.062418. arXiv: 2401.07716.

(7) "Estimating quantum mutual information through a quantum neural network,"
Myeongjin Shin, Junseo Lee, and Kabgyun Jeong,
Quantum Information Processing **23**, 57 (2024). DOI: 10.1007/s11128-023-04253-1. arXiv: 2306.14566.

(6) "Quantum Rényi entropy functionals for bosonic gaussian systems,"
Junseo Lee, and Kabgyun Jeong,
Physics Letters A **490**, 129183 (2023). DOI: 10.1016/j.physleta.2023.129183. arXiv: 2204.10737.

(5) "Weighted p-Rényi entropy power inequality: Information theory to quantum Shannon theory,"
Junseo Lee, Hyeonjun Yeo, and Kabgyun Jeong,
International Journal of Theoretical Physics **62**, 253 (2023). DOI: 10.1007/s10773-023-05512-8. arXiv: 2311.06484.

(4) "High-dimensional private quantum channels and regular polytopes,"
Junseo Lee, and Kabgyun Jeong,
Communications in Physics **31**, 189 (2021). DOI: 10.15625/0868-3166/15762. arXiv: 2101.00230.

(3) "Single qubit private quantum channels and 3-dimensional regular polyhedra,"
Kabgyun Jeong, Junseo Lee, *et al.*,
New Physics: Sae Mulli **68**, 232 (2018). DOI: 10.3938/NPSM.68.232
Bronze Award, The Humantech Paper Award, Samsung Electronics (2018).

Book Chapters

(2) "Assessing Quantum Integer Factorization Performance with Shor's Algorithm,"
Junseo Lee, In: *Quantum Computing: A Journey into the Next Frontier of Information and Communication Security*,
Edited by Mohammad Hammoudeh, Abdullah T. Alessa, Amro M. Sherbeeni, Clinton M. Firth, Abdullah S. Alessa,
CRC Press (2024). DOI: 10.1201/9781003475286

Patents

(1) "Method for estimating quantum mutual information through a quantum neural network," Kabgyun Jeong, Myeongjin Shin, and Junseo Lee, *Korea Patent* Open No. 10-2026-0009068 (2024). DOI: 10.8080/1020240091151

Selected Honors and Awards

Scholarships and Fellowships

- Kwanjeong Ph.D. Fellowship ($25,000/year), *Kwanjeong Foundation*, 2026–2030
- QSE Professional Development Fellowship ($4,000), *Harvard University*, 2026
- Academic Travel Grant ($3,000), *Hyundai Motor CMK Foundation*, 2022
- Science and Technology Scholarship (full tuition and stipend), *Hyundai Motor CMK Foundation*, 2021–2022
- Teaching Excellence Scholarship, *Yonsei University*, 2021–2022

Academic Excellence

- High Honors, *Yonsei University*, 2022
- Honors, *Yonsei University*, 2020–2021
- Science Scholarship, *JH Foundation*, 2017
- Academic Excellence Scholarship, *Asan Future Scholarship Foundation*, 2016

Additional Honors and Awards

- Selected Paper Award, Finance and Economics Contest, *DB Cultural Foundation*, 2022
- Best Tutor Award, *Yonsei University*, 2021–2022
- Third Prize, Undergraduate Research Exhibition, *Korean Physical Society*, 2021
- Bronze Award, Humantech Paper Award, *Samsung Electronics*, 2018
- Best Translator Award, *NAVER Connect Foundation & Khan Academy*, 2018
- Pre-Undergraduate Research Fellowship, *Korea Advanced Institute of Science and Technology*, 2017–2018
- Honorable Mention (National) & Gold Award (Regional), Korean Olympiad in Informatics (KOI), 2016

Professional Activities

Journal Reviewer

- **American Physical Society:** Physical Review Letters, PRX Quantum, Physical Review Research, Physical Review Applied, Physical Review A
- **Nature Portfolio:** Nature Communications, npj Quantum Information
- **Institute of Electrical and Electronics Engineers:** IEEE Transactions on Information Theory
- **Others:** Quantum, Physics Letters A, Annalen der Physik, Mathematical Methods in the Applied Sciences, Statistics and Probability Letters, Algorithmica, Journal of Circuits, Systems, and Computers

Conference Reviewer

- Quantum Information Processing (QIP), 2027
- ACM-SIAM Symposium on Discrete Algorithms (SODA), 2027
- IEEE Symposium on Foundations of Computer Science (FOCS), 2026
- Theory of Quantum Computation, Communication and Cryptography (TQC), 2026
- Quantum Computing Theory in Practice (QCTiP), 2026
- Quantum Techniques in Machine Learning (QTML), 2025

Organizing

- Co-organizer, Harvard Quantum Computing Theory Group Seminar, 2027–Present
- Organizer, Winter School on Quantum Learning Theory for Bosonic and Fermionic Systems, 2026
- Co-organizer, Quantum Information Theory Seminar, *Seoul National University*, 2024–2025

- Co-organizer, Quantum AI Hackathon, *Kakao Enterprise & Jeonju University*, 2025

Committee Service

- Poster Session Committee Member, National Undergraduate Quantum Conference, *Seoul National University*, 2026
- Selection Committee Member, Quantum Internship Program, *National Information Society Agency & Korea Quantum Industry Center*, 2024–2025

Outreach and Community Resources

- Creator and Maintainer, Quantum Learning Theory Zoo, 2025–Present
- Facilitator (Mentor), Korea Scholar's Conference for Youth (KSCY), *Yonsei University*, 2019

Teaching

*Teaching materials available at: harris-junseo-lee.github.io/teaching/.

Harvard University & Massachusetts Institute of Technology (2027–)

- Teaching Fellow, MIT 6.5430/Harvard CS 2232: Frontiers of Quantum Complexity Theory†, Fall 2027 (Graduate Course)
- Teaching Fellow, Harvard Physics 260A/QSE 210A: Introduction to Quantum Information I, Spring 2027 (Graduate Course)

Special Lecture Series, Quantum Information Science Club Association (2025–2026)

- Organizer and Instructor, Quantum Learning Theory for Bosonic and Fermionic Systems*, Winter 2026
- Instructor, Quantum Complexity Reading Group*, Fall 2025
- Instructor, Quantum Learning and Complexity Theory*, Summer 2025

University-Industry Research Internship (2024–2025)

- (External) Instructor, AAA559: Internship II, *Korea University*, Fall 2025 (Graduate Course)
- (External) Instructor, AAA558: Internship I, *Korea University*, Fall 2025 (Graduate Course)
- (External) Instructor, SW4343: Software Field Placement I, *Korea Aerospace University*, Fall 2024

Yonsei University (2021–2022)

- Teaching Assistant, YCS1009: Change the World through Programming, Fall 2022
- Teaching Assistant, YCS1002: Software Programming, Fall 2022
- Teaching Assistant, EEE1108: Engineering Information Processing, Fall 2021
- Course Tutor, MAT2016: Engineering Mathematics III, Spring 2022 (Best Tutor Award)
- Course Tutor, MAT1012: Engineering Mathematics II, Fall 2021 (Best Tutor Award)

Research Visits

- Harvard University (supported by Prof. Anurag Anshu and the Harvard Quantum Initiative), Mar. 2026

Selected Talks

*Online talk.

Research Talks

"Heisenberg-limited Hamiltonian learning without short-time control"

- Invited talk, *Technische Universität Hamburg (Hamburg University of Technology)*, Jun. 2026*
- Long contributed talk, 26th Asian Quantum Information Science Conference (**AQIS 2026**), *Korea Advanced Institute of Science and Technology (KAIST)*, Aug. 2026

"Certifying and learning local quantum Hamiltonians"

- Invited talk, Quantum Meets (Online Seminar Series), Jun. 2026*
- Invited talk, 7th QNRC Joint Workshop, *Korea Institute of Science and Technology Information (KISTI)*, May 2026
- Contributed talk, 21st Conference on the Theory of Quantum Computation, Communication and Cryptography (**TQC 2026**), *Delta Hotels Sherbrooke Conference Centre (Québec, Canada)*, Sep. 2026

"Optimal certification of constant-local Hamiltonians"

- Invited talk, Quantum Software Lab Seminar, *University of Edinburgh*, Mar. 2026*

- Contributed talk, Annual Meeting of the Quantum Information Society of Korea (QISK 2026), *Seoul SETEC Convention Center*, Feb. 2026

"Efficient learning of bosonic Gaussian unitaries"

- Invited talk, Annual Meeting of the Quantum Information Society of Korea (QISK 2026), *Seoul SETEC Convention Center*, Feb. 2026
- Invited talk, N^{3}etFraST Workshop, *Korea Institute of Science and Technology Information (KISTI)*, Nov. 2025
- Invited talk, Quantum Data Science & AI Lab Seminar, *Yonsei University*, Nov. 2025
- Contributed talk, 29th Annual Conference on Quantum Information Processing (**QIP 2026**), *University of Latvia*, Jan. 2026 (presented under the title "Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators")

"Resource-efficient algorithm for estimating the trace of quantum state powers"

- Invited talk, *Electronics & Telecommunications Research Institute (ETRI)*, Dec. 2024
- Invited talk, Quantum Information Theory Seminar, *Seoul National University*, Dec. 2024*
- Invited talk, IBM-Yonsei Qiskit Fall Fest, *Yonsei University*, Nov. 2024*
- Invited talk, KISTI-KU-SNU Joint Workshop, *Seoul Biohub*, Oct. 2024
- Contributed talk, Annual Meeting of the Korean Mathematical Society (KMS), *Sungkyunkwan University*, Oct. 2024
- Poster, 28th Annual Conference on Quantum Information Processing (QIP 2025), *Raleigh Convention Center (NC, United States)*, Feb. 2025

"Mutual information maximizing quantum generative adversarial network"

- Invited talk, Triangle Quantum Computing Seminar, *NC State University Quantum Initiative*, Nov. 2023*

"Estimating quantum mutual information through a quantum neural network"

- Invited talk, CS Katha Barta, *National Institute of Science Education and Research (Bhubaneswar, India)*, Aug. 2023*

"Quantum Rényi entropy functionals for bosonic Gaussian systems"

- Poster, 25th Annual Conference on Quantum Information Processing (QIP 2022), *Caltech*, Mar. 2022

"High-dimensional private quantum channels and regular polytopes"

- Invited talk, KISTI-KU-SNU Joint Workshop, Sep. 2023*
- Invited talk, Quantum Information Theory Seminar, *Seoul National University*, Aug. 2021*
- Contributed talk, Winter Meeting of the Optical Society of Korea (OSK), *Daejeon Convention Center*, Feb. 2022
- Contributed talk, Fall Meeting of the Korean Physical Society (KPS), Oct. 2021*
- Poster, 25th Annual Conference on Quantum Information Processing (QIP 2022), *Caltech*, Mar. 2022

Tutorials and Public Lectures

"Getting Started in Quantum Information Science"

- Invited talk, Student Outreach Lecture, *Shinil High School (Seoul, Korea)*, Apr. 2026
- Invited talk, Student Outreach Lecture, *Shinil High School (Seoul, Korea)*, Aug. 2024

"Learning theory in ∞-dimensional quantum systems"

- Invited talk, Team QST Summer Workshop, *Seoul National University*, Aug. 2025

"Introduction to quantum machine learning"

- Invited talk, Healthcare & Research Team Seminar, *Amazon Web Services (AWS Seoul, Korea)*, Mar. 2025

"Quantum machine learning models for drug library generation"

- Invited talk, Quantum Computing and Monte Carlo Workshop, *Yonsei University*, Aug. 2024

"$\mathsf{QMA} \stackrel{?}{=} \mathsf{NP}$: The NLTS theorem and the quantum PCP conjecture"

- Invited talk, Center for Quantum Network's Channel Capacity Summer Workshop, *Seoul National University*, Jul. 2024

"Minimal data may be sufficient for quantum artificial intelligence"

- Invited talk, Department of Mathematical Sciences Seminar, *Seoul National University*, Jun. 2023*